CAI4201: Natural Language Processing
7th Assignment

CoNLL-2003(https://www.kaggle.com/datasets/alaakhaled/conll003-englishversion) is a benchmark dataset for Named Entity Recognition (NER), originally introduced for the Conference on Computational Natural Language Learning (CoNLL) 2003 shared task. It contains English newswire text from the Reuters corpus, manually annotated for named entities

The task is to identify and classify mentions of four entity types:

- **PER** — persons, e.g., "Bill Clinton"
- **ORG** — organizations, e.g., "Microsoft"
- **LOC** — locations, e.g., "Japan"
- **MISC** — miscellaneous named entities that do not fit the other categories, such as nationalities, events, or products.

The English dataset is divided into three standard splits:

Split	**Number of sentences**
Training	14,041
Validation	3,250
Test	3,453

Each sentence is stored one token per line, with an empty line separating sentences. In the standard files, each token has four space-separated columns: the word, its part-of-speech tag, its syntactic chunk tag, and its named-entity tag.

Annotation format

The named-entity labels use the **IOB2/BIO** scheme:

- `O` means the token is not part of a named entity.
- `B-TYPE` marks the beginning of an entity of type `PER`, `ORG`, `LOC`, or `MISC`.
- `I-TYPE` marks a token inside an entity.

For example, in "New York City," `New` may be labeled `B-LOC`, while `York` and `City` are labeled `I-LOC`. The `B-` tag is used when two entities of the same type occur immediately next to each other and the second one must be distinguished as a new entity.

Assignment task

Students should build models that assign an NER tag to every token in a sentence. After prediction, adjacent `B-` and `I-` tags are reconstructed into complete entity spans and compared with the gold annotations.

The primary evaluation metric should be span-based entity-level F1: a predicted entity is correct only when both its boundaries and its entity type exactly match a gold entity. Students should also report precision, recall, and per-class results for PER, LOC, ORG, and MISC.

Suitable experiments

This dataset is appropriate for comparing feature-based and embedding-based NER approaches, including:

- N-gram models or n-gram features for tag prediction.
- Word2Vec embeddings with classifier of your choice and dicuss this choice.
- GloVe embeddings with a classifier of your choice and dicuss this choice.
- LSTM classifier using N-gram Word2vec and GloVE as input features
- LSTM classifer but using input embedding layer instead of perviouse features

Students can additionally investigate whether lexical context, capitalization, part-of-speech tags, and chunk tags improve performance. The dataset's inclusion of POS and chunk annotations makes it particularly useful for controlled feature ablation experiments.

Submission Requirments

1- All of your python implmentations.
2- A report that dicuss the results of the expirments from your point of view.
3- Upload a record on youtube that explains the implmentation and the results analysis and submit the link.

Note: AI generated reports or copies will get a 0